TECHNICAL SPECIFICATIONS

DETAILS OF ITEMS FOR COLLEGE OF MILITARY ENGINEERING

Note :-
1. Appropriate original technical literature / pamphlets supporting the specifications should be provided.

2. Failure to provide the requisite information such as make and model will make the Bid null & void and will not be consider further.

3. Technical specifications as have been asked should be replied with "Yes" or "No" against each item in the "specifications of the item being provided" column and not your own specification be proposed. If higher specifications are available with the Firm a separate comparative statement of specification as have been asked for by this Institution and being offered by the Firm can be attached. Offering your own specification in Tech bid will reject your bid automatically.

Upgradation of Trg Infrastructure with Latest Bomb Disposal and Counter IED Eqpt

Ser No	Specification desired by the Faculty	Make and Model No	Specifications of item accepted by the vendor
1	**MINI CALIBER ROBOT: -** (i) **Technical Parameter** (aa) Generation – 2nd Generation Claw attachment (ab) Camera – PTZ (ac) Video Recorder – Digital upto 320GB (ad) LCD – 13 cm Touch Screen (ii) **Robot Body** (aa) Materials - Aircraft-grade aluminium alloy precision machining. (ab) Dimensions - >L*W*H: 910 * 650 * 500 mm. (ac) Weight –≥70kgs(without accessories, package and control box). (ad) Battery - DC24V lead acid rechargeable battery. (ae) Working time - DC24V lead acid rechargeable battery. (af) Working time - ≥ 5 hours (ag) Maximum Speed - ≥7km/h (ah) Loading Capacity - When loading 100KG, it can move normally (actual measurement). (aj) Lift Capability - It can move with clamping weights of 20Kgs and will not drop (actual measurement). (ak) Grade Ability - It can climb up the slope of 45° and stop steadily on the slope. (al) Climbing Stairs Ability - With traction-free assist, it can climb up and down the stairs of 200mm step height and 45°angle slope. (am) Drag Capability - ≥100kgs (an) Turning Ability - In horizontal cement ground or bituminous pavement, the robot can turn clockwise or anticlockwise 360º. (ao) Limited Passage Width - ≤700mm (ap) Over-obstacle Capacity - It can cross the obstacle of 320mm height. (aq) Max. Mechanical Arms Spread - 1650mm (ar) Gripper of Manipulator Maximum Expansion Range-250mm (as) Arm Extension when stretch out and draw back – 500mm		

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	(at) Control Box - Via the buttons and handle of control box, the operators can control video switch, zoom, moving, cradle rotating, floodlight, telescopic arm, gripper opening and closing, small arm rotating, big arm lifting, kidney basin rotating, wrist rotating, etc. (au) 6 Mechanical Arm DOFs - Manipulator gripper opening and closing. Manipulator gripper rotation. Small arm rotation. (av) Rotation Angle - Manipulator gripper can rotate clockwise and anticlockwise 360º.Small arm can rotate clockwise 150º and anticlockwise 110º.Big arm can pitch angle of 90º.Kidney basin can rotate clockwise 180º and anticlockwise 45º.Control Distance - Wireless control: ≥200m (visible range); wire control: 100m (optional 200m); (iii) **<u>Number of Cameras and Floodlight</u>** (aa) Forward Camera - Colour infrared induction (ab) Backward Camera - Colour infrared induction (ac) Cradle Head Various focal Camera - Colour infrared induction. (ad) Manipulator Gripper Camera - Colour infrared induction (ae) Floodlight - Two group LED floodlight (one group on the front and back). (ae) Thermal imaging camera in PTZ. (iv) **<u>Monitoring cradle head</u>** (aa) DOFs of Cradle Head (ab) 2 independent DOFs, adjustable rotation speed. (ac) Cradle rotation: 360º continuously, maximum rotation speed of 130° per second. (ad) Cradle pitching: -45°~ 120°, maximum pitching speed of 50° per second. (v) **<u>Control Terminal</u>** (aa) Box Portable, waterproof, dustproof, high strength (ab) Size ≤ L 460 * W 370 *H 260 mm (ac) Weight ≤ 10kg (ad) Display Screen 12-inch HB LCD, wide viewing angle, outdoor clear picture Operation. (ae) High-quality rocker handle, human software interface design, easy observation and convenient operation (af) It can monitor 4 video signals simultaneously or separately amplify one of 4video signals (ag) Information Display Real-time voltage (ah) Battery Rechargeable 24V lithium battery, working time ≥ 5.1 hours when fully charged. (vi) **<u>Additional attachments</u>** (aa) X Ray mount (ab) Water disrupter mount (ac) Extension Arm (ad) Weapon mount (ae) Motorised anti flip		

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2	**<u>EXPLOSIVE VAPOUR DETECTOR</u>** (i) **<u>Technology</u>**- FLIR True Trace, multi-channel fluorescence technology, no radioactive source. (ii) **<u>Sampling &Analysis</u>**: - (aa) <u>Sample Introduction</u> - Direct vapor for bottled liquids orsampling swipes for surfaces vapor and trace particulate. (ab <u>Sample Phase</u> - Vapor and trace particulate. (ac) <u>Threats</u> - Detects military conventional, homemade and liquid explosives. (ad) <u>Sampling & Analysis</u> - Analysis≤10 seconds with real time detection capability. (iii) **<u>System Interface</u>**: - (aa) <u>Display & Alerts</u> - Visible, audible and haptic (vibration) alerts, vivid sunlight readable color display (32k colors). (ab) <u>Localization</u>: - Language translation available worldwide. (ac) <u>Communication</u>: - USB; optical embedded wireless; window based software. (ad) <u>Data storage</u>:- 3500 hrs of continuous data logging. (ae) <u>Data Management</u>: - On device report generation; export to mass storage. (af) <u>User Levels</u>: - Operator and Administrator. (iv) **<u>Power</u>**: - (aa) <u>Input Voltage</u>: - 100-240V AC (wall adapter supplied). (ab) <u>Battery Spaces</u>: - Two Rechargeable Li-on batteries (swappable and up to 8hrs battery life); adapter. (ac) <u>Startup Time</u>: - <5 minutes from cold;<10 second from sleep. (v) **<u>Environmental</u>**:- (aa) <u>Operating Temperature</u>: - 14 to 122^{o}F (-10to 50^{o}C). (ab) <u>Operating Humidity: -</u> 5% to 95% non-condensing. (ac) <u>Storage Temperature</u>: - -4 to 122 ^{o}F (-20 to 50^{o}C) (vi) **<u>Physical Features</u>**: - (aa) <u>Dimension (LxWxH)</u>:- 14.5x4.5x2.8 in (370x11.5x7.0cm) with battery. (ab) <u>Weight</u>:- ≥3.0 lbs (1.4kg). (ac) <u>Enclosure & Protection</u>:- Molded magnesium and polymer composite casing with anticorrosive coating. (ad) <u>Standards</u>:- ASTM ^{TM} E2520. (viI) **<u>Detection capability</u>**: - (aa) All type of service and commercial explosive can detect. (ab) Also can detect homemade explosive (ac) library can be edit on field. (ad) provide various type of chemical/explosives		

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3.	**ROCKET WRENCH** (i) The equipment should be provided with various optional claws, which can be fitted and used as required depending on the shape of the fuze to be withdrawn. (ii) It should be light in weight, rugged and portable design. (iii) It should be made of corrosion resistance, all weather proof and non- sparking material. (iv) The design of Rocket wrench should be accurate and efficient. (v) The Venturies and Breech plug should be replaceable. (vi) Following additional accessories are required along with Rocket Wrench assembly as a spare. (aa) qty-02 Venturies (ab) qty-02 Breech Plug Elect (ac) qty-02 Breech plug (ad) qty-02 Dust Cap (ae) qty-01 each different types of jaws (af) qty-02 Bungee card of good quality and strength (vii) It should have the capability to give high torque during operation. (viii) Equipment should be able to function at extreme ranges of temperatures i.e. -20 degree C to +50 degree C. (ix) Equipment should be provided with carrying case with hard foam inset. (x) It should be provided with cleaning brush and spare kit. (xi) The Jaws should be adjustable as per requirement. (xii) Hardened and strengthened Spanners should be provided for tightening of Jaws and opening of Breech.		
4.	**DIGITAL ELECTRONIC STETHOSCOPE/ ELECTRONIC CONTACTLESS STETHOSCOPE** (i) It is designed for specific operations in the field search, IEDD and EOD and is mainly used for the detection of weak sound waves such as clockwork timers (Mechanical, electromagnetically and Digital), a common component of IEDs or incendiary devices as well as radio proximity fuzes with active command decoders. (ii) **Amplifier** (aa) LED for battery monitoring. (ab) Adjuster for amplification with rotary ON/ OFF switch. (ac) Step less Filter- approx. 2-6 KHz. (ad) Sockets for Headset and Probes. (ae) Rechargeable battery 9 Volt/ 160 mA, Approx. 06 hrs operation time in single charge. (af) Battery Charging time- Approx. 14 hr with 11 mA. (ag) Maximum Amplification- approx. 7 x 10^5 (117 dB). (ah) Temperature ranges- -20 to +55--60^0C. (aj) MPS Sensor capacity- Around 10-15 cm (iii) **Contactless Probe** (aa) Requires its own built-in rechargeable Battery 12 V/ 600 mA. approx. operation time min 06 hours in single charge.		

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	(ab) Charging time- approx. 14 hours with 50 mA. (ac) LED- for indication of Battery voltage. (ad) Exchangeable fuse. (ae) Detection range- More than 30 cm. (iv) **<u>Transit case</u>** Shock proof transit/ storage case.		
5.	**<u>DE ARMOUR DISRUPTOR</u>: -** (i) It should be developed as a dual role equipment to defeat the threat posed by both Improvised Explosive Devices (IEDs) and unexploded Ordnance (UXO). (ii) It should have equipped with the supplied disruption barrel the equipment will project a high pressure water slug into the suspect device disrupting the firing circuit and providing a high probability of avoiding detonation. (iii) Disruptor should be made from corrosion resistant materials. (iv) Dimension should be 58 cm x 38 cm x 53 cm. (v) Weight of main unit should be 21.4 kg. (vi) Shooting Speed should be 600 m/s. (vii) It should not fall or back off after shooting. (viii) Shooting Energy should be 13000J. (ix) It should be powered by 9V battery. (x) Tube diameter should be 38 mm. (xi) Provide 100 Nos cartridge.		
6.	**<u>BACK SCATTER X-RAY MACHINE</u>** (i) Lead detection and imaging capability. (ii) Depth in Detection capability via Vide Ray Infinity Detector and Optics. No need to imaging panel. (iii) Most compact scanner dimensions where it matters. (iv) Remote connectivity to any device (PC, MAC, iOS and Android). (v) Large 7” Diagonal 700-nit bright color touch screen display. (vi) Simple Over-The-Air (OTA) Updates (vii) Line Laser to better outline the scan area. (viii) X-ray Source minimum 150 keV (ix) Weight less than 5 kgs (x) Operating System Software: Android, current version (xi) Multi-Language User Interface capable (xii) Total System sys Weight for tpt less than 14kgs (xiii) Operating Temperature: -200^{o} C (-4^{o}F) to 60^{o} C (140^{o} F) (xiv) Storage Temperature: -40^{o}C (-40^{o}F) to 70^{o} C (158 ^{o}F) (xv) Operating Conditions: Rain, snow, high winds, and altitudes up to 4,500 m (14,764 ft) nominally. (xvi) Fully CE compliant (xvii) <u>Radiation safety</u>: Complies fully with all applicable U.S. federal health and safety regulations, Meets ANSI, ICRP, NCRP, and Erratum radiation safety standards for annual allowable dose for the general public, to help prevent inadvertent X-ray emission, the system is equipped with a series of interlocks and visual indicators, Contains no live radiation source (xviii) Steel Penetration: up to 25mm of steel. (xix) Scanning speed Up to 30 cm per second; 15cm per second (Nominal Scan Speed for a 1:1 Scan Image to Scan Target Object Ratio)		

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	(xxi) Display -OLED. (xxii) Storage Capacity: min 4 GB. (xxiii) Pixel Pitch: 140µm. (xxiv) Active Area: (355.6*430)/ (385*460) I 83.6%. (xxv) Active Pixels: 2560X3072 (7.8M pixels). (xxvi) Scintillator: CsI. (xxvii) Resolution: 3.57lp/mm. (xxviii) Image Bit Depth: 16bit PNG. (xxix) Data Interface: Gigabit Ethernet / Wireless (802.11.ac). (xxx) Calibration Mode: ACC (Automatic Calibration Control)/Manual. (xxxi) Data Transfer Time: Less than 1 sec. (xxxii) Dimension [WXLXH]: 385X460X15mm. (xxxiii) Voltage: 18V D.C./0.5A. (xxxiv) **Battery** (aa) Capacity: 14.8V D.C./4100mAh/58.4Wh, 2 x btys. (ab) Charging: 16.78V/0.5A. (ac) Life Time: minimum 5 hours. (ad) Type: Li-ion Polymer Battery. (xxxv) **System Components** (aa) PX1 scanner (ab) Rugged transport cade (ac) Two rechargeable batteries (ad) Battery charger (ae) Data transfer cable (af) Shoulder strap (ag) Two wrist straps (ah) Quick Start Guide		
7.	**Triple sensor Metal detector**:- (i) It is designed for detection of magnetic and non-magnetic in all soil condition, Command wire detection, Enhanced IED Detection of targets such as electrical conductive, non-magnetic devices, Short wires and high resistivity metals. (ii) **Search Head** (aa) Large scan width. (ab) Narrow front size. (ac) Detection in all directions. (ad) Minimum Setup time. (ae) Integrated GPS and data logging system. (af) Command wire detection capability. (ag) Detection magnetic and non-magnetic in all soil condition. (ah) IED Detection of targets such as electrical conductive, non-magnetic devices, Short wires and high resistivity metals. (aj) Detection range- Min 1 Mtr. (iii) **Control Unit** (aa) Integrated battery with charger. (ab) Battery level monitoring. (ac) Separate acoustic, visual and tactile alarm signalling of each class of target. (ad) Built in speaker. (ae) Built in vibration signalling indicator.		

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	(af) Built in wireless earphone. (ag) Easy to operate. (ah) Ambient light sensor. (iv) High immunity against external interference. (v) Compact and light weight design. (vii) Superior adaption to all type of soils. (viii) Easy to carry. (ix) All weather robust and tough design. (x) One-piece foldable design with carrying strap. (xi) Visual signalling selectable as visible or infrared compatible with night vision device. (xii) Operating temperature between -30^{o} to +60^{o} C		
8.	**<u>LIQUID EXPLOSIVE DETECTOR</u>** (i) LED gives a precise result since it automatically finds the optimal focus distance to the sample, even inside containers. It can differentiate, between the container and the actual substance, giving improved and faster results during investigation of Post Blast incidents of Level-I categories. ii) **<u>Explosive</u>** Includes military and civilian explosive, IED's precursors and oxidants. (iii) **<u>Narcotics</u>** Narcotic and psychotropic substance from the UN Yellow and Green lists including pre cursors. (iv) **<u>Hazardous Chemicals</u>** Contains more than 2200TICs (Toxic Industrial Chemicals) and other toxic hazardous materials. (v) **<u>Chemical Warfare Agents</u>** Blister, blood, never and choking agents and their precursors. vi) **<u>Specifications</u>** (a) Instrument type- Hand Held (b) Laser excitation wavelength-785nm (c) Barcode Reader – Yes QR and most common barcodes formats. (d) laser output power – 3 levels, max 300mW (e) Max spectral range- 400cm-1 to 2, 300cm-1 (f) Auto-focus – 0-4mm (g) Spectral resolution- 8 to 10cm-1 (h) Detector type – Linear CCD Array (j) Display – 3.5'' transflective TFT with LED backlight. (k) Memory – More than 100000 measurements (l) Data formats - .txt, csv, .jcamp (m) Connectivity – USB C, Wi-fi (2.4GHz) (n) Battery – Rechargeable more than 12 hours of typical use (o) Weight – not more than 1 kg. (p) Size – as per weight (q) Operating temp - -20^{o} to +50^{o}C. (r) Storage temp - -30^{o} to +50^{o}C. (s) Included accessories- Vial holder, watertight case, protective cap with calibration target, sample vials, USB- C –Cable, AC Adapter 5 VDC/1 A USB for charging. (t) Optional accessories- SERS kit to identify low concentration and fluorescent samples.		

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	(u) Start Up time-less than 20 sec in room temperature. (v) Scan time- Most substances less than 10 sec (w) Exposure – Auto exposure or manual exposure, 0.001- 60 sec. (x) Optional Libraries – Explosive, include precursors, narcotics including cutting agent, masking agent and precursors, Chemical warfare agents, including simulants, hazardous material, including industrial chemicals, Pharmaceuticals with a wide range of active pharmaceutical ingredients. Total number of substances is more than 14000. (y) Warranty- 5 years. Additional warranty option available. (z) MTBF - 50000 h. (aa) Relative humidity- 5 to 90% (ab) Certificates- CE, IEC 60529 IP67, MIL-STD-810G 514.6 (vibration), MIL-STD-810G 516.6 (shock), 21 CFR Part11 (optional), SWGDRUG category A compliant for presumptive testing. (ac) Supported languages- English (ad) Software included- ChemDash Lite, Optional software, Chemdash Pro. (ae) Type of substances- Solid, liquids, powders and tablets. (af) Mixture analysis- Yes (ag) Delay function- Yes, 1 to 15 minutes. (ah) Features- Auto-Focus, Quickscan, User authorisation, Event logging, Scan delay, User-defined libraries, User-added substances.		
9.	**CHEMICAL ANALYSER** (i) Utilizes both Raman spectroscopy & FTIR technology. (ii) Provides complementary and confirmatory results ideal for advanced users. (iii) General characterization of chemicals. (iv) HazMasterG3 Software (add on capability). (v) Determine potential hazardous material being made from on site chemicals. (vi) Determine what HME formula ingredients are missing. (vii) **Screener Mode** (available with LowDoseID) (aa) Utilizes Raman and SERS (ab) Ideal for all users levels (ac) Narcotics Identification (viii) **Direct Scan** (aa) Analysis via holder or probe. (ab) Substances that are pure or in high concentration. (ix) **Type H Scan** (aa) Substances present in low concentration (ab) Items with fluorescence issues (x) Size-10.1 in x 5.7 in x 2.4 in (25.6 cm x 14.6 cm x 6.1 cm). (xi) Weight- 4.2 lbs (1.9 kg) (xii) Spectral range - FTIR: 4,000 cm^{-1} to 650 cm^{-1}Raman: (785nm) 250 cm^{-1} to 2875 cm^{-1}.		

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	(xiii) Spectral resolution- FTIR: 4 cm^{-1} Raman: 7 to 10.5 cm^{-1} (FWHM) across range. (xiv) Collection optics- FTIR: ATR diamond crystal, single bounce. (xv) Raman: Fiber optic probe (xvi) Power supply- Internal 3.7V lithium polymer (QTY 1) battery pack Internal CR123A (QTY 3) batteries DC wall adapter, 12 V DC 1.25 A (xvii) SERS-1.5 ml solvent vial via polycarbonate H-stick with substrate. (xviii) Data export- SPC (for use in standard spectroscopic software), reach back (.rbk), text, or PDF (xix) Ruggedness- MIL-STD-810G and IP67 (xx) Languages- English, German, Turkish, Spanish, Chinese, Arabic, French, Russian (xxi) Add on capability- Optional HazMasterG3® decision support software (English GUI only); requires Gemini v1.6 or later software. Low dose ID for low concentration analysis. On-board mixture analysis- Identification of up to 4 components in a mixture.		
10	**<u>DRONE WITH WATER DISRUPTER</u>** **(i) Ground Control Station(GCS) Features** (aa) Ruggedized/ Commercial Laptop/Tablet (ab) Pre-Flight Self-checks (ac) Return Home on Communication Failure (ad) Return Home on Imbalance (ae) Return Home on GPS failure **(ii) Ground Control Station(GCS) Software** (aa) Integrated Application software with geographical mapsand video view with single click switch ability between the two. (ab) Real-time video streaming from UAV with on-screendisplay of important parameters (ac) Beyond Line of Sight Capability (ad) Multiple UAV Management (ae) Third Party Parameter Integration **(iii) Technical Specifications (Model Specific)** (aa) Endurance:> 30 mins @ MSL (ab) Range (Line of Sight): 2-3 km (ac) Payload weight: <10 Kg (ad) Wind Resistance: 36 km / hr **(iv) Payloads(360 pan & 90 tilt)** (aa) Daylight Payload Resolution: 1920 x 1080 pixels 10x optical zoom (ab) HD Imaging Payload: 16 MP(Upgradable) (ac) Thermal Payload Resolution: 640 x 480 Pixels Black Hot/ White Hot/ InstAlert/Fusion Modes (ad) Megaphone Capability (ae) Dust/ Water ingress rating: IP 56 (af) Vehicle Size: 150 cm x 150 cm (ag) Maximum Launch Altitude: 3000m AMSL (ae) Maximum Operating Altitude: 500m AGL **(v) Fail Safe Features** (aa) Multiple GPS for redundancy (ab) Return Home on Low Battery		

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	(ac) Return home on Communication Failure (ad) Return home on Imbalance (vi) Application Areas (aa) Counter Insurgency &Surveillance (ab) Crowd Monitoring (ac) Search & Rescue (ad) Aerial Patrol (ae) Medical First Response (af) Large area monitoring of Lands/ Mines (ag) Equipment Delivery (ah) GIS Mapping (aj) Crop Monitoring & Spraying (ak) Utility Infrastructure Inspection (al) Can attch water disrupter (am) Endurance 40 mins minimum (an) 2 x bty pack		
11	**<u>DRONE WITH SEARCH AND EXPLOSIVE DETECTION CAPABILITY</u>** (i) **<u>GPR and MagnetometersystemSpecifications</u>** **The system can be used for both airborne and terrestrial surveys.** (ii) **<u>ANTENNA / GPR</u>** (aa) Type: Single channel Ground Penetrating Radar system with shielded antenna. (ab) Centre frequency: 1000 MHz. (ac) Operating bandwidth: 600–1300 MHz (-6 dB). (ad) Sampling: Real Time Sampling (RTS) with high hardware stacking. (ae) Samples per scan: 256 / 512 / 1024 / 2048 / 4096 / 8192. (af) Samples rate: up to 1,280,000 (depends on hardware stacking). (ag) Scan rate: up to 2,500 scans/s (depends on hardware stacking). (ah) Sample output: 32-bit digital raw data. (aj) Time range per scan: 18 ps / 36 ps / 71 ps / 89 ps / 125 ps / 143 ps / 179 ps / 250 ps / 357 ps / 500 ps / 625 ps / 714 ps / 1 ns / 1.25 ns / 2.5 ns (ak) Depth: Up to 2 m depending on ground properties. (al) Data format: Standard Geophysical SEGY (am) Data Format (.sgy) with traces geotagging. (an) Vertical filters: Off, Digital. (ao) Horizontal filters: Stacking, Background removal. (ap) Gain points: 1–10 Linear gain. (aq) Gain levels: 0 to +168 dB. (ar) Data storage: Raw data storage with memory for Gain used. (as) Data file size: Limited only by available storage space. (iii) **<u>ENVIRONMENTAL</u>** (aa) Temperature: -20°C to 60°C internal temperature (ab) Humidity: 96% non-condensing. (ac) Ingress protection.		

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	(ad) IP68 in protective box (used for transportation and ground surveys) (ae) IP52 in airborne variant. (iv) **MECHANICAL (Weight):** (aa) 1.8 kg – airborne configuration without mounting kit. (ab) 7.0 kg – in protective transportation box. (ac) 10.0 kg – in carton box for delivery. (v) **BATTERY** (aa) Airborne variant: Drone main battery powers the GPR. (ab) Ground surveys: Battery integrated in protective box, operating time up to 12 hours. (ac) 99 W/h Li-Ion2 x btys. (vi) **Components** (aa) Zond Aero 1000 NG GPR system with 1000 MHz center frequency antenna. (ab) Prism 2 Data Acquisition/Processing software (ac) NANUK transport case (also used for ground surveys) (ad) 2 x 99 W/h Li-Ion battery integrated into the case (used for ground surveys only). (ad) Wi-Fi router integrated into the case (used for ground surveys only). (ae) Mascot charger (af) Tow rope for terrestrial surveys (ag) Mounting kit to fix GPR system/ Magnetometer on drone		
12	**EOD 10E (NIJ)** (i) The EOD is to give best-in-class balanced protection and support to bomb technicians. Every layer of protective material was carefully selected for optimum performance to weight ratio. (ii) It transfers the weight from shoulders to the waist to reduce fatigue and improve comfort, increases situational awareness, provides upper body ventilation and adds new capabilities. (iii) The helmet is equipped with a unique inflatable bladder and multi-point retention system for a secure and customized fit that shapes closely to each user's head. This eliminates the need for additional fit pads. (iv) All protective materials were evaluated and carefully patterned to reduce as much weight as possible. The results are outstanding: a lighter ensemble that reduces operator fatigue. (v) The improved ergonomic design allows superior range of motion for the arms, legsand waist, and permits the head to look around more freely. (vi) Users will immediately feel a noticeable improvement in mobility due to significantweight reduction, ergonomic design and the use of more advanced materials. (vii) A patented system enables emergency extraction by robot or manually. Severaldesign features have been optimized to facilitate faster and simpler access todeliver medical aid.		

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	(viii) **MISSION-SPECIFIC LIGHTS**: Users can choice of 3 LED lights: White, Red or Blue. Their physical properties enable the user to select the best option for the mission environment, such as tactical scenarios, reducing risk when a photosensitive device is suspected, or maximizing illumination of a dark space. (ix) **SITUATIONAL AWARENESS** : The development program emphasized the need to help technicians interact with their environment. The system includes stereophonic acoustics to determine the directional source of sounds, digital speakers, warnings and alerts, new lights and an increased field of view. (x) **VOICE COMMAND SYSTEM** :The EOD 10E includes a built-in Voice Command System that allows a user to control the helmet and suit's many electronically activated features while using their hands to carry tools or use Hook & Line. This helps complete the mission more efficiently during manual tasks (xii) **REMOTE CONTROL UNIT**: The EOD 10E Remote Control Unit is a significant advancement over the previous generation. It controls an array of new features and uses icons and numbers to convey information clearly and precisely. It is operated by voice commands or buttons that are raised and backlit. (xiii) **SYSTEM CONFIGURATION**: The Remote Control Unit can configure and save several functions, such as audio settings, ventilation and lights. The system can save up to 4 sets of user preferences to serve multiple team members or be preset for different operational environments. (xiv) **CONFIRMATIONS, WARNINGS &DIAGNOSTICS**: The helmet uses coloured lights inside the visor and audio signals to convey mission-critical confirmations and warnings, such as a low power level. In addition, the Remote Control Unit displays troubleshooting guidance to keep the system in service. (xv) **INTEGRATED COOLING**: The EOD mitigates heat stress through two new ventilation systems: one in the jacket, the other in the helmet. The flow rates in these two systems can be controlled individually or simultaneously. Ambient air can also be chilled by the addition of optional cooling accessories. (xvi) **OVERPRESSURE**: The EOD provides superior protection against both the blast overpressure and impulse associated with explosive threats. In addition, the EOD Helmet offers significant improvements for ear overpressure protection against closer proximity blasts. (xvii) **FRAGMENTATION**: The suit and helmet have been extensively tested against 17, 44 and 207 grain fragment simulated projectiles, by independent NIJ- approved test facilities. This range of fragment sizes and shapes better represents random fragments projected by an IED. Advancements in ballistic materials and processing techniques have yielded higher V50 protection levels in the helmet and suit while reducing armour weight. (xviii) **IMPACT**: A completely redesigned helmet protection system for the head and back areas provides industry-leading impact protection as tested in accordance with latest standards.		

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	This includes testing in extreme heat and cold conditions to better represent operational climates worldwide. (xix) **<u>HEAT:</u>** The outer shell materials of the suit and helmet provide excellent flame protection, including after flame, afterglow, flaming melt drip, and char length, as well as flash heat from an explosive event. (xx) **<u>NIJ</u>** The EOD is officially certified to the US National Institute of Justice NIJ 0117.01 standard for Public Safety Bomb Suits. In addition, it offers additional enhancements in functionality and uses the latest technologies to provide the modern bomb technician with new capabilities and enhanced user interfaces. (xxi) **<u>EMC COMPATIBILITY</u>**: The increasing threat of Radio Controlled IEDs means a bomb suit must be compatible with Electronic Countermeasures (ECM). The EOD filters and shields its systems to suppress unwanted signals from being emitted, and prevents unwanted incoming signals from reaching its electronics. To ensure this, it conforms to the exacting electronic and electrical standards of MIL-STD-461E using a variety of frequency ranges. (xxii) **<u>CHEM/BIO PROTECTION</u>**: The EOD can be equipped with an optional Breathing Apparatus Visor that is compatible with many commonly- used respirators. (xxiii) **<u>FOOT PROTECTION</u>**: The Med-Eng engineering team has developed a new Foot Protection System that provides 360° protections and allows a much improved natural walking movement. A ratchet system secures it to the user's boot for a customized fit and makes it almost unnoticeable when worn. (xxiv) Design The certified EOD Bomb Suit Ensemble (baseline configuration) includes: (aa) EOD Suit: ▪ Jacket ▪ Front Jacket Panel ▪ Neck/Chest/Groin Plate Package ▪ Rear Jacket Panel with Multi-Connector Cable (MCC) and Back Protector with Integrated Ventilation ▪ One-Piece Trousers ▪ Trouser Expansions (ab) EOD Helmet: ▪ EOD Visor (Face Shield and transparent Visor) with Built-In Lights ▪ Inflatable and Adjustable Impact and Sizing Liner ▪ Four-point Retention System ▪ Visor Cover for transportation and storage ▪ 3 x Balaclavas (ac) Primary Electronics Module (PEM) and Power Supply (ad) Wrist-mounted Remote Control Unit (RCU) (ae) Grounding Straps for Electrostatic Discharge (ESD) (af) User Manual, User Video, Suit & Helmet Donning Guides and RCU Quick Start Guide		

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	(ag) Suit Hanger (ah) Carry Bags supplied with both the Suit and Helmet (aj) Foot Protection (Toe Protectors or Full Foot Protectors) (aj) 2x bty reqd for all electronic units and bty charger also available in local markets.		
	Warranty Period. Minimum three years.		
	Installation and Training. The OEM/Dealer should install the equipment at the desired location and conduct demonstration and training for handlers.		
	AMC - AMC should be provided 4 years after completion of warranty period (i.e 3 years). Bidder should be quoted AMC amount on their letter head in percentage form. The AMC rates will not be taken as consideration for deciding L1 vendor. AMC contract will be placed separately. If vendor does not quote for AMC clause while bidding will be rejected directly at TEC stage.		

(Stamp and Signature of Vendor)